

PARK RAPIDS, MN, USA

WMO: 727453

Lat: 46.901N

Lon: 95.068W

Elev: 1434

StdP: 13.95

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94967

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.6	-17.9	-27.8	1.2	-22.1	-23.4	1.6	-16.8	26.8	14.7	24.4	14.4	4.9	300	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	87.4	70.8	84.1	68.8	81.6	67.2	74.0	83.4	71.7	80.2	69.8	78.4	13.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.2	121.3	79.6	68.9	112.1	77.2	66.5	102.9	75.0	38.5	83.1	36.4	80.1	34.7	78.5	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.5	21.0	19.0	DB	-31.3	92.6	7.8	2.6	-36.9	94.4	-41.4	96.0	-45.7	97.5	-51.4	99.4	(4)
			WB	-29.5	77.5	5.7	1.9	-33.6	78.9	-37.0	80.0	-40.1	81.1	-44.3	82.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	40.6	8.1	12.8	25.9	41.1	54.0	63.8	69.0	66.6	58.1	43.9	28.3	14.3	(6)
		DBStd	23.70	14.71	13.56	13.38	10.32	8.72	6.68	5.96	5.94	8.70	9.12	11.59	13.15	(7)
		HDD50	5468	1300	1043	750	296	58	2	0	25	235	652	1107		(8)
		HDD65	9258	1765	1463	1212	716	355	98	30	51	240	656	1100	1572	(9)
		CDD50	2047	0	0	3	30	181	415	589	515	266	45	3	0	(10)
		CDD65	363	0	0	0	1	14	61	154	101	31	1	0	0	(11)
		CDH74	3567	0	0	0	15	206	624	1458	936	302	26	0	0	(12)
		CDH80	929	0	0	0	2	50	137	425	239	72	4	0	0	(13)
		WSAvg	8.9	8.7	8.7	9.1	9.6	9.6	8.7	7.9	7.9	8.8	9.5	9.1	8.7	(14)
(15)	Precipitation	PrecAvg	25.3	0.7	0.5	1.1	2.2	2.7	4.2	3.7	3.6	2.9	2.4	1.0	0.7	(15)
		PrecMax	40.5	2.0	1.6	2.9	5.9	6.7	9.0	11.6	9.5	5.7	6.5	3.8	1.9	(16)
		PrecMin	12.5	0.0	0.0	0.1	0.2	0.2	0.7	0.4	0.3	0.4	0.2	0.1	0.2	(17)
		PrecStd	4.5	0.4	0.3	0.6	1.2	1.6	2.0	2.2	2.3	1.5	1.8	0.7	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	45.9	64.0	75.2	85.8	88.8	92.1	90.5	87.0	77.9	63.0	43.5	(19)
			MCWB	35.8	40.0	53.0	56.6	65.2	72.5	73.7	72.6	69.0	59.9	52.2	38.7	(20)
		2%	DB	34.8	39.9	55.8	69.1	79.5	83.8	87.7	85.5	81.6	69.9	55.3	37.3	(21)
			MCWB	32.1	35.5	46.6	51.7	61.5	68.6	72.2	69.7	67.4	57.5	47.1	34.2	(22)
		5%	DB	31.7	36.1	48.7	64.2	74.7	80.6	84.4	82.2	77.2	63.5	48.8	34.1	(23)
			MCWB	29.9	33.1	42.0	48.9	58.5	65.0	69.9	67.7	64.4	54.0	42.6	32.1	(24)
		10%	DB	28.2	32.5	43.0	59.2	70.3	77.6	81.8	79.5	72.9	58.6	44.0	31.4	(25)
			MCWB	26.4	30.3	37.8	46.3	56.1	63.6	68.5	65.9	62.6	50.4	39.2	29.7	(26)
		0.4%	WB	36.1	40.4	56.0	58.9	68.8	75.3	77.3	75.9	72.6	63.4	54.4	39.3	(27)
			MCDB	40.2	45.4	60.8	73.4	80.9	85.5	86.0	86.2	82.2	69.2	59.0	43.6	(28)
		2%	WB	32.5	36.3	47.1	53.8	64.6	71.3	74.8	72.4	69.6	59.0	47.7	34.6	(29)
			MCDB	34.5	39.2	54.7	64.3	74.2	80.0	84.2	81.1	78.6	68.0	53.4	37.0	(30)
(31)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	29.8	33.2	42.2	50.4	61.4	68.5	72.7	70.4	66.9	54.7	43.4	32.3	(31)
			MCDB	31.7	35.9	48.5	62.0	70.9	76.8	81.5	78.6	73.8	62.2	48.8	34.1	(32)
		10%	WB	26.6	30.3	38.2	47.4	58.3	66.1	70.5	68.3	64.2	51.3	39.4	29.7	(33)
			MCDB	28.1	32.6	42.9	58.0	67.2	74.0	78.8	76.0	71.0	56.9	43.8	31.3	(34)
		5% DB	MDBR	19.4	20.6	19.6	21.2	21.9	20.4	21.2	21.4	20.8	18.0	15.9	16.2	(35)
			MCDBR	17.6	19.7	24.1	30.3	29.2	25.0	23.8	24.9	25.7	26.9	23.2	17.4	(36)
(37)	Mean Daily Temperature Range	5% WB	MCWBR	15.3	15.5	15.8	16.6	14.9	12.1	11.4	11.6	13.1	15.7	15.2	14.6	(37)
			MCDBR	17.3	17.9	23.0	26.5	23.8	20.3	21.6	21.2	20.8	23.7	21.8	16.2	(38)
		5% WB	MCWBR	15.2	14.4	15.5	15.8	13.6	12.0	12.0	11.3	12.2	14.9	15.1	13.9	(39)
			MCDBR	17.3	17.9	23.0	26.5	23.8	20.3	21.6	21.2	20.8	23.7	21.8	16.2	(40)
(41)	Clear-Sky Solar Irradiance	taub	0.266	0.305	0.302	0.354	0.370	0.375	0.378	0.380	0.351	0.307	0.280	0.261		(41)
		taud	2.337	2.214	2.390	2.339	2.356	2.367	2.379	2.388	2.451	2.557	2.518	2.400		(42)
		Ebn at Noon	264	271	294	286	284	282	279	274	273	273	261	254		(43)
		Edh at Noon	23	32	32	37	38	37	37	35	30	23	19	19		(44)
(44)	All-Sky Solar Radiation	RadAvg	504	863	1217	1534	1700	1856	1976	1663	1234	758	462	351		(44)
		RadStd	55	87	122	131	116	127	89	137	92	118	47	43		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	-2.39	-2.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page